

Nahid Ul Islam, Ph.D.

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[Webpage](#) | [LinkedIn](#) | [Google Scholar](#) | [GitHub](#)

SUMMARY

- Ph.D. in Computer Science at Arizona State University (ASU); developed foundation models for large-scale medical image analysis in collaboration with **Mayo Clinic**; published **6 conference and 2 journal papers** with **170+ citations** with **h-index 7**.
- AI/ML Researcher and Engineer specializing in **Computer Vision** and **Deep Learning** with expertise in **foundation model pretraining, multi-task/multimodal learning, Swin Transformer, and Vision Transformers** for large-scale image analysis.
- Inventor of **3 granted U.S. patents** and recipient of the **ASU President's Award for Innovation** in AI and Deep Learning research.
- Experienced in **model development**: large-scale experiment design, benchmarking, and optimization across diverse imaging tasks.
- Completed **2 industry internships** in Machine Learning and Computer Vision at **Intel Corporation**.
- Proficient in **Python, MATLAB, C/C++, and Java**, with expertise in deep learning frameworks such as **PyTorch** and **TensorFlow**.

EDUCATION

- **Doctor of Philosophy (Ph.D.) in Computer Science** 2018 – 2026
Arizona State University, USA
GPA: 4.00/4.00
Research: Computer Vision, Deep Learning, Foundation Models
- **Master of Science (M.Sc.) in Computer Science** 2015 – 2017
University of Texas at San Antonio, USA
GPA: 3.90/4.00
Research: Computer Vision, Machine Learning, Image Processing
- **Bachelor of Science (B.Sc.) in Computer Science** 2010 – 2014
BRAC University, Bangladesh
Research: Image Processing

WORK EXPERIENCE

Arizona State University

Tempe, AZ

Graduate Research Assistant (*ASU-Mayo Clinic Joint Research Collaboration*)

Aug 2018 – May 2026

- Published **8 peer-reviewed papers** (6 conference, 2 journal) on Computer Vision, Deep Learning and Medical Image Analysis.
- Inventor of **3 granted and 5 pending U.S. patents** in AI and Deep Learning, developed in collaboration with Mayo Clinic.
- Built scalable **AI/ML frameworks** on **High-Performance Computing (HPC) GPU infrastructure** using distributed training, Linux/SLURM-based remote computing, workflow orchestration, and large-scale data pipelines for production-scale AI systems.
- Developed a systematic benchmark comparing 12 CNN architectures, all major Vision Transformer and Swin Transformer variants, and 19 self-supervised pretrained models for pulmonary embolism diagnosis; proposed **Embedding-based ViT (EViT)**, a two-stage framework that first learns 2D slice-level representations and then aggregates them for 3D exam-level diagnosis.
- Designed **Foundation X/X+**, a multi-task foundation model for chest X-ray analysis integrating image classification, object detection (bounding box), and segmentation for disease analysis; conducted large-scale pretraining across 11 datasets and 20 tasks (~826k images); introduced Lock-Release Pretraining to stabilize multi-task optimization and reduce task-specific overfitting; validated through extensive quantitative analysis demonstrating strong cross-task and cross-dataset generalization.
- Developed **Foundation CTPA/Chest CT-CTPA**, a unified end-to-end slice-based 3D foundation model integrating classification, lesion detection, and segmentation within a single framework, pretrained on over 1.5 million CT slices for coherent 2D slice-level and 3D volume-level learning; introduced Region-Guided ROI Alignment to transfer lesion detection cues to segmentation; demonstrated strong generalization to unseen downstream benchmarks and cross-domain transferability to CT imaging tasks.

Intel Corporation (*Client Computing Group*)

Hillsboro, OR

Graduate Technical Intern — Machine Learning and Computer Vision

May 2017 – Aug 2017

- Benchmarked R-CNN, Fast R-CNN, Faster R-CNN, and Mask R-CNN for real-time obstacle detection, classification, and collision prediction for moving objects.
- Evaluated model performance and presented experimental findings and actionable recommendations to the team, contributing to pathfinding and technical readiness decisions.

Intel Corporation (*Client Computing Group*)

Hillsboro, OR

Graduate Software Engineering Intern — Machine Learning and Computer Vision

Feb 2017 – May 2017

- Researched deep learning approaches for human activity recognition from videos to enable smart home applications.
- Developed an activity recognition system combining CNNs and optical flow for temporal modeling; conducted iterative experiments to improve model accuracy and robustness.

TECHNICAL SKILLS

- **Programming Languages**
Python, C/C++, Java, MATLAB, Bash/Shell
- **Machine Learning Framework/Library**
PyTorch, TensorFlow, Hugging Face Transformers, Scikit-Learn, OpenCV, NumPy, Pandas, Weights & Biases, Git/GitHub
- **Computing & Infrastructure**
High-Performance Computing (HPC), Distributed Training, Linux, SLURM, Large-Scale Data Pipelines
- **Core Artificial Intelligence (AI) & Machine Learning (ML)**
Deep Learning, Foundation Models, Large-scale Pretraining, Transfer Learning, Fine-tuning, Multitask/Multimodal Learning, Supervised/Self-Supervised Learning, CNNs, Large Language Models (LLMs), Transformer Architectures (Vision/Swin)
- **Model Development, Experimentation & Analysis**
Experiment Design, Ablation Studies, Statistical Significance Testing, Benchmarking, Qualitative/Quantitative Analysis, Explainable AI, Attention/Embedding Analysis (Grad-CAM/t-SNE), Model Monitoring/Debugging
- **Expanding Skills** [\[GitHub\]](#)
Retrieval-Augmented Generation (RAG), LangChain, Vector Databases (FAISS), Prompt Engineering, REST API Development (FastAPI), Containerization (Docker)

PUBLICATIONS

[[Google Scholar](#): Citations 170+ | h-index 7]

Peer-refereed Journal Publications

1. **Islam, N. U.**, Ma, D., Gotway, M. B., & Liang, J. (2026). *Foundation Chest CT-CTPA: Maximizing the Utility of Heterogeneous Expert Annotations via a Unified Slice-Based 2D-3D Foundation Model*. Medical Image Analysis. *[under review]*
2. **Islam, N. U.**, Ma, D., Pang, J., Gotway, M. B., & Liang, J. (2026) *Foundation X⁺: Enhanced Knowledge Sharing for Integrated Classification, Localization, and Segmentation with Lock-Release Pretraining and Region-Guided ROI Alignment in Chest X-ray Analysis*. Medical Image Analysis. *[under review]*
3. **Islam, N. U.**, Zhou, Z., Gehlot, S., Gotway, M. B., & Liang, J. (2024). *Seeking an optimal approach for Computer-aided Diagnosis of Pulmonary Embolism*. Medical Image Analysis, 91, 102988.
4. Guo, Z., **Islam, N. U.**, Gotway, M. B., & Liang, J. (2024). *Stepwise incremental pretraining for integrating discriminative, restorative, and adversarial learning*. Medical Image Analysis, 103159.

Peer-refereed Conference Publications

1. **Islam, N. U.**, Ma, D., Gotway, M. B., & Liang, J. (2026). *A Generic 3D Foundation Model Utilizing 2D and 3D Annotations for Classification, Localization, and Segmentation in CT Pulmonary Angiography*. In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops, Large Foundation Models in Biology and Biomedicine (LFMBio 2026).
2. **Islam, N. U.**, Ma, D., Pang, J., Senthil Velan, S., Gotway, M. B., & Liang, J. (2025). *Foundation X: Integrating Classification, Localization, and Segmentation through Lock-Release Pretraining Strategy for Chest X-ray Analysis*. In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV).
3. **Islam, N. U.**, Gehlot, S., Zhou, Z., Gotway, M. B., & Liang, J. (2021). *Seeking an optimal approach for computer-aided pulmonary embolism detection*. In Machine Learning in Medical Imaging: 12th International Workshop, MLMI 2021, Held in Conjunction with MICCAI 2021, Strasbourg, France, September 27, 2021, Proceedings 12 (pp. 692-702). Springer International Publishing.
4. Guo, Z., **Islam, N. U.**, Gotway, M. B., & Liang, J. (2022, September). *Discriminative, restorative, and adversarial learning: Stepwise incremental pretraining*. In MICCAI Workshop on Domain Adaptation and Representation Transfer (pp. 66-76). Cham: Springer Nature Switzerland.
5. Pang, J., Haghighi, F., Ma, D., **Islam, N. U.**, Hosseinzadeh Taher, M. R., Gotway, M. B., & Liang, J. (2022, September). *POPAR: Patch Order Prediction and Appearance Recovery for Self-supervised Medical Image Analysis*. In MICCAI Workshop on Domain Adaptation and Representation Transfer (pp. 77-87). Cham: Springer Nature Switzerland.
6. Ma, D., Hosseinzadeh Taher, M. R., Pang, J., **Islam, N. U.**, Haghighi, F., Gotway, M. B., & Liang, J. (2022, September). *Benchmarking and boosting transformers for medical image classification*. In MICCAI Workshop on Domain Adaptation and Representation Transfer (pp. 12-22). Cham: Springer Nature Switzerland.

Publicly Released Software (Selected)

1. Foundation X/X+, an end-to-end model integrating classification, localization, and segmentation tasks [\[GitHub\]](#)
2. Foundation CTPA/Chest CT-CTPA, a slice-based 3D foundation model for chest CT imaging [\[GitHub\]](#)
3. Evaluating and optimizing deep learning methods for computer-aided diagnosis of pulmonary embolism [\[GitHub\]](#)
4. Benchmarking and boosting transformers for medical image classification [\[GitHub\]](#)
5. Restoring patch order and appearance for Self-supervised Medical Image Analysis [\[GitHub\]](#)

Granted

- Nahid UI Islam**, Shiv Gehlot, Zongwei Zhou, Jianming Liang. U.S. Patent No. 12,236,592, “Systems, methods, and apparatuses for systematically determining an optimal approach for the computer-aided diagnosis of a pulmonary embolism” issued Feb 25, 2025.
- Zuwei Guo, **Nahid UI Islam**, Jianming Liang. U.S. Patent No. 12,572,813, “Systems, methods, and apparatuses for implementing discriminative, restorative, and adversarial (DiRA) learning using stepwise incremental pre-training for medical image analysis” issued March 10, 2026.
- DongAo Ma, Jiaxuan Pang, **Nahid UI Islam**, Mohammad Reza Hosseinzadeh Taher, Fatemeh Haghighi, Jianming Liang. U.S. Patent No. 12,394,186, “*Systems, methods, and apparatuses for implementing self-supervised domain-adaptive pre-training via a transformer for use with medical image classification*” issued Aug 19, 2025.

Pending (Selected)

- Nahid UI Islam**, DongAo Ma, Nathaniel Alberti, Jianming Liang. U.S. Provisional Patent Application No. 63/921,066, “A Generic 3D Foundation Model Utilizing 2D and 3D Annotations for Classification, Localization, and Segmentation in CT Pulmonary Angiography” filed November 19, 2025.
- Nahid UI Islam**, DongAo Ma, Jiaxuan Pang, Shivasakthi Senthil Velan, Jianming Liang. U.S. Provisional Patent Application No. 63/744,744, “*Foundation X+: Enhanced Knowledge Sharing for Integrated Classification, Localization, and Segmentation with Lock-Release Pretraining and Region-Guided ROI Alignment in Chest X-ray Analysis*” filed Jan 13, 2025.
- Nahid UI Islam**, Zongwei Zhou, Shiv Gehlot, Jianming Liang. U.S. Non-Provisional Patent Application No. 18/904,995, “*Systems, Methods, and Apparatuses for Computer-Aided Diagnosis of Pulmonary Embolism*” filed Oct 2, 2024.
- Zuwei Guo, **Nahid UI Islam**, Jianming Liang. U.S. Patent Application No. US20250029372A1, “*Systems, methods, and apparatuses for implementing stepwise incremental pre-training for integrating discriminative, restorative, and adversarial learning into an AI model*” filed July 17, 2024.
- Jiaxuan Pang, Fatemeh Haghighi, DongAo Ma, **Nahid UI Islam**, Mohammad Reza Hosseinzadeh Taher, Jianming Liang. U.S. Patent Application No. US20240078666A1, “*Systems, methods, and apparatuses for implementing patch order prediction and appearance recovery (popar) based image processing for self-supervised learning medical image analysis*” filed Sept 1, 2023.

AWARDS AND HONORS

- Earned **SUN Award**, Arizona State University, for invited research talk and providing subject matter expert feedback Apr 2026
- Received the prestigious **President’s Award for Innovation**, Arizona State University
Project Title: ‘*Annotation-efficient Deep Learning for Computer-aided Diagnosis in Medical Imaging*’ Dec 2024
- Awarded prestigious **Doctoral Fellowship** at Arizona State University Aug 2018
- Honored by **VC’s List Award** for securing highest GPA (4.0 out of 4.0) in the last semester at BRAC University Dec 2014

THESIS

- Arizona State University, Ph.D.**
Maximizing the Utility of Expert Annotations for Medical Image Analysis
- University of Texas at San Antonio, M.Sc.**
Face Detection and Age-Gender Estimation Using Patch-Based DCT and HOG Features
- BRAC University, B.Sc.**
Automated Parking Lot Management via License Plate and Bengali Text Recognition Using Image Processing

TALKS AND PRESENTATIONS

- Presented GPU utilization for Deep Learning and Computer Vision, guiding peers on workflows for large-scale training.
- Delivered findings from published research in graduate-level seminars, including:
 - Seeking an Optimal Approach for Computer-Aided Diagnosis of Pulmonary Embolism*
 - Foundation X: Leveraging Annotations for Comprehensive Multi-Task Learning in Chest X-ray Analysis*
 - Foundation CTPA: A Generic 3D Model Utilizing 2D and 3D Annotations for Comprehensive Chest Imaging Tasks*
- Presented *Learning Unified Chest Radiograph Representations from Heterogeneous Expert Models and Annotations* at BIDS Seminar (ASU–Mayo)

SERVICES, LEADERSHIP AND TEAM-WORK

Served as a **reviewer** for leading venues including MICCAI, WACV, MIDL, Media, CVPR, and TPAMI, and as a **Travel Grant Reviewer** at ASU; held elected **leadership roles** in student organizations at ASU and UTSA, and received recognition for award-winning cultural performance at BRAC University.